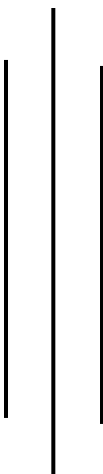


TRIBHUWAN UNIVERSITY

CENTRAL DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY



OBJECT ORIENTED SOFTWARE ENGINEERING

Assignment No-1

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Assignment 1

Create a program illustrating Classes, Objects, Instances, Polymorphism, Inheritance and Encapsulation.

Program

```
# Parent class
class Animal:
    """Parent Class accepts name and sound"""
    def __init__(self, name, sound):
        self.name = name    # Public attribute
        self._species = "Unknown" # Protected attribute
        self.__sound = sound # Private attribute

    def make_sound(self):
        return f'{self.name} makes a {self.__sound} sound.'

    def get_species(self):
        return self._species # Accessing protected attribute

# Child class (Inheritance)
class Dog(Animal):
    """Child Class takes Dog name and breed"""
    def __init__(self, name, breed):
        super().__init__(name, "bark") # Calling parent constructor for sound
        self._species = "Canine" # Modifying protected attribute for species
        self.breed = breed # Dog Class new attribute

# Overriding method (Polymorphism)
def make_sound(self):
    return f'{self.name} ({self.breed}) barks.'

# Another child class (Polymorphism)
class Cat(Animal):
    """Child class takes Cat name"""
    def __init__(self, name):
        super().__init__(name, "meow")
        self._species = "Feline"

    def make_sound(self):
        return f'{self.name} meows.'

# Creating objects of Dog and Cat Class
```

```

dog = Dog("Buddy", "Golden Retriever")
cat = Cat("Whiskers")

# Accessing public, protected, and private data
print(dog.make_sound()) # Polymorphism
print(cat.make_sound()) # Polymorphism

print(f'{dog.name} belongs to species: {dog.get_species()}') # Accessing protected attribute
print(f'{cat.name} belongs to species: {cat.get_species()}')

# Encapsulation (trying to access private variable directly)
try:
    print(dog.__sound)
except AttributeError:
    print("Cannot access private attribute '__sound' directly! Encapsulation")

```

Output

```

PS C:\Users\Yujan\OneDrive\Desktop\Master\OOSE> & C:/Python312/python.exe c:/Users/Yujan/OneDrive/Desktop/Master/OOSE/OOP_Python.py
Ghost (Golden Retriever) barks.
Mighty meows.
Ghost belongs to species: Canine
Mighty belongs to species: Feline
Cannot access private attribute '__sound' directly! Encapsulation

```